

Mathematics for Computing

Assignment - Report & Module 3

Anlin Davis

M.Tech AI & SE

A Report on Official Statistical Machinery

? Prepare a report on the activities and functions of the Department of Statistics under the Government of Kerala and the Government of India, with a maximum length of 500 words.

Official Statistics - Introduction

- Statistics play an important role in government administration, economic planning, policy formulation and evaluation of development programmes.
- In India, statistical activities are carried out at both the State and Central levels. The Department of Economics and Statistics (DES) is the nodal agency for official statistics in Kerala, while the Ministry of Statistics and Programme Implementation (MoSPI) is the nodal ministry for the development and coordination of the statistical system at the national level.

Department of Economics and Statistics, Government of Kerala

- The Department of Economics and Statistics is responsible for the systematic collection, compilation, analysis, interpretation and dissemination of statistical information relating to Kerala's economy. It provides reliable data required for planning and evidence-based decision-making.
- Major activities of the department include the collection of demographic and vital statistics, agricultural statistics, industrial statistics, price statistics, labour and housing statistics, and state income estimates. It conducts sample surveys, agricultural surveys, the Annual Survey of Industries and various ad-hoc surveys and studies. The department also prepares estimates of Gross State Domestic Product (GSDP) and district income.
- The department publishes important statistical reports such as the Statistical Handbook, Panchayat Level Statistics, Gender Statistics, Infrastructure Statistics and Environmental Statistics. It also provides statistical dashboards and data portals for public access. The department has a Directorate, 14 District Statistical Offices, 61 Taluk Statistical Offices and statistical cells in various government departments.

Statistical System of the Government of India

- At the national level, the Ministry of Statistics and Programme Implementation (MoSPI) coordinates and develops India's official statistical system. Its Statistics Wing, functioning through the National Statistical Office (NSO), conducts large-scale surveys and produces important national statistics.
- Major activities include the Periodic Labour Force Survey (PLFS), Household Consumption Expenditure Survey (HCES), Annual Survey of Industries (ASI), Annual Survey of Unincorporated Sector Enterprises (ASUSE), and other socio-economic surveys. MoSPI also compiles National Accounts Statistics, including Gross Domestic Product (GDP), and important price statistics such as the Consumer Price Index (CPI) and Index of Industrial Production (IIP).
- MoSPI develops the National Indicator Framework for monitoring Sustainable Development Goals (SDGs). Its Programme Implementation Wing monitors major central sector projects and the Members of Parliament Local Area Development Scheme (MPLADS).

Conclusion

Thus, statistical departments at both State and Central levels provide essential data for planning, resource allocation, policy formulation and evaluation. Kerala's Department of Economics and Statistics focuses on state-level statistical requirements, while MoSPI coordinates and produces national-level statistics. Together, they support evidence-based governance and socio-economic development.



Department of Economics and Statistics, Government of Kerala,
<https://www.ecostat.kerala.gov.in/>



Ministry of Statistics and Programme Implementation, Government of India,
<https://www.mospi.gov.in/>

Module 3

Descriptive Statistics

Descriptive statistics is the branch of statistics concerned with collecting, organizing, summarizing and presenting data in a meaningful way. It helps to transform large amounts of raw data into a form that is easier to understand and interpret. It uses numerical measures such as the mean, variance, and covariance as well as graphical methods such as histograms, to summarize the important characteristics of data.

The major objectives are:

- To describe the central tendency of data.
- To measure the variability or spread of data.
- To understand the distribution and pattern of observations.
- To identify relationships between different variables.
- To present data using tables, graphs and numerical measures.

Histogram

A **histogram** is a graphical representation of the distribution of numerical data.

The data is divided into intervals called **bins** or **class intervals**. The frequency of observations in each interval is represented by the height of a bar.

Suppose the marks of 10 students are:

45, 50, 52, 55, 60, 62, 65, 70, 75, 80

The data can be grouped as follows:

Marks	Frequency
40–49	1
50–59	3
60–69	3
70–79	2
80–89	1

A histogram represents these frequencies using adjacent rectangular bars.

Sample Mean

The **sample mean** is the arithmetic average of all observations in a sample. The sample mean is often used to estimate the population mean μ . Let the sample be

$$X_1, X_2, \dots, X_n.$$

The sample mean is given by

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

• where : $\bar{X}$ - sample mean , X_i - i th observation , n - sample size.

Consider the sample:

$$10, 20, 30, 40, 50$$

$n = 5$, Therefore,

$$\bar{X} = \frac{10 + 20 + 30 + 40 + 50}{5} = 30$$

Sample Variance

Sample variance measures how much the observations in a sample vary around the sample mean. The sample variance is

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

Consider the sample:

10, 20, 30

sample mean $\bar{X} = 20$.

The deviations from the mean are

$$10 - 20 = -10, 20 - 20 = 0, 30 - 20 = 10.$$

The squared deviations are

100, 0, 100.

Therefore,

$$S^2 = \frac{100 + 0 + 100}{3 - 1} = 100$$

Order Statistics

Order statistics are the observations of a sample arranged in increasing or decreasing order. Suppose the original sample is

7, 2, 9, 4, 5.

After arranging the observations in ascending order:

2, 4, 5, 7, 9.

The ordered observations are denoted by

$$X_{(1)}, X_{(2)}, \dots, X_{(n)}.$$

Here,

- $X_{(1)}$ - smallest observation, $X_{(n)}$ - largest observation.

The minimum is

$$X_{(1)}$$

and the maximum is

$$X_{(n)}.$$

For an odd number of observations, the median is

$$X_{(\frac{n+1}{2})}$$

Consider

3, 8, 2, 10, 5.

Ordered data:

2, 3, 5, 8, 10.

Therefore:

- Minimum = 2
- Median = 5
- Maximum = 10

Sample Covariance

Covariance measures the direction of the relationship between two variables.

Suppose we have two variables

$$X = (X_1, X_2, \dots, X_n)$$

and

$$Y = (Y_1, Y_2, \dots, Y_n).$$

The sample covariance is

$$S_{XY} = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})$$

where $\bar{X}$ and $\bar{Y}$ are the sample means.

Covariance

Positive

Negative

Approximately

Interpretation

X and Y tend to increase together.

When one variable increases, the other tends to decrease.

There is little or no linear relationship.

Sample Covariance Example

Suppose:

	X	Y
1	1	2
2	2	4
3	3	6

As X increases, Y also increases.

Therefore, the covariance is positive.

Note: Covariance indicates the direction of a relationship, but its magnitude depends on the units of measurement.

Sample Covariance Matrix

When there are more than two variables, their variances and covariances can be represented using a **sample covariance matrix**.

For three variables X , Y , and Z , the covariance matrix is

$$S = \begin{bmatrix} S_X^2 & S_{XY} & S_{XZ} \\ S_{YX} & S_Y^2 & S_{YZ} \\ S_{ZX} & S_{ZY} & S_Z^2 \end{bmatrix}$$

The diagonal elements represent variances, while the off-diagonal elements represent covariances.

Consider the covariance matrix

$$S = \begin{bmatrix} 4 & 2 & 1 \\ 2 & 9 & 3 \\ 1 & 3 & 16 \end{bmatrix}.$$

Here:

- Variance of $X = 4$
- Variance of $Y = 9$
- Variance of $Z = 16$
- Covariance between X and $Y = 2$
- Covariance between X and $Z = 1$
- Covariance between Y and $Z = 3$

A covariance matrix is:

- Square
- Symmetric
- Positive semi-definite
- Characterized by variances on its diagonal
- Characterized by covariances off its diagonal

Frequentist Statistics

Frequentist statistics is an approach to statistical inference that uses sample data to make conclusions about unknown characteristics of a population. In this approach, population parameters such as the mean and variance are considered fixed but unknown quantities, while the sample data and statistics calculated from the sample are treated as random.

The main purpose is to estimate population parameters, test statistical hypotheses and quantify the uncertainty associated with statistical estimates.

The major objectives are:

- To obtain information about a population from sample data.
- To measure the accuracy and reliability of estimators.
- To construct confidence intervals for population parameters.
- To study the properties of statistical estimators.
- To make decisions and conclusions based on sample evidence.

Sampling

Sampling is the process of selecting a subset of observations from a larger population.

- A **population** is the complete group of individuals or observations being studied.
- A **sample** is a smaller group selected from the population.

Suppose a college has 5,000 students. If 200 students are selected for a survey:

- Population = 5,000 students
- Sample = 200 students

The objective is to use the sample information to draw conclusions about the population.

1. Simple Random Sampling

Every member of the population has an equal probability of being selected.

2. Stratified Sampling

The population is divided into groups called strata, and samples are taken from each stratum.

3. Systematic Sampling

Individuals are selected at regular intervals, such as every k th individual.

4. Cluster Sampling

The population is divided into clusters, and some clusters are selected for study.

Mean Square Error

Mean Square Error (MSE) measures the average squared difference between an estimator and the true population parameter. For an estimator $\hat{\theta}$ of a parameter θ :

$$MSE(\hat{\theta}) = E[(\hat{\theta} - \theta)^2]$$

MSE can be decomposed into variance and squared bias:

$$MSE(\hat{\theta}) = \text{Var}(\hat{\theta}) + [\text{Bias}(\hat{\theta})]^2$$

where

$$\text{Bias}(\hat{\theta}) = E[\hat{\theta}] - \theta.$$

Suppose an estimator has

$$\text{Var}(\hat{\theta}) = 4 \text{ and } \text{Bias}(\hat{\theta}) = 2.$$

Then

$$MSE = 4 + 2^2 = 8$$

A smaller MSE indicates that is closer to the true parameter in squared-error terms.

Consistency

An estimator is called **consistent** if it gets closer to the true population parameter as the sample size increases.

Suppose $\hat{\theta}_n$ is an estimator of θ . The estimator is consistent if

$$\hat{\theta}_n \xrightarrow{P} \theta$$

as

$$n \rightarrow \infty.$$

This means that the probability that the estimator is far from the true parameter becomes very small as the sample size increases.

The sample mean $\bar{X}$ is a consistent estimator of the population mean μ under standard conditions.

More data	$\implies$	Better estimate	$\implies$	Closer to the true value
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Confidence Intervals

A **confidence interval (CI)** provides a range of plausible values for an unknown population parameter. In general,

$$\text{Confidence Interval} = \text{Estimate} \pm \text{Margin of Error}$$

Suppose the estimated population mean is

$$\bar{X} = 50$$

and the 95% confidence interval is

$$(46, 54).$$

The interval is reported as : $46 < \mu < 54$ using the stated 95% confidence procedure.

- 90% , 95% , 99%
- A higher confidence level generally produces a wider confidence interval, assuming the other factors remain unchanged.
- A 95% confidence procedure means that if samples were repeatedly taken and confidence intervals were constructed using the same method, approximately 95% of those intervals would contain the true population parameter.

Parametric Model Estimation

A **parametric model** assumes that the population distribution belongs to a particular family described by a finite number of parameters. A normal distribution can be written as

$$X \sim N(\mu, \sigma^2)$$

where : μ - population mean , σ^2 is the population variance.

The objective of estimation is to estimate these unknown parameters using sample data.

- Normal distribution , Binomial distribution , Poisson distribution and Exponential distribution

Maximum Likelihood Estimation (MLE) chooses the parameter value that maximizes the likelihood of observing the given data.

The likelihood function is

$$L(\theta) = \prod_{i=1}^n f(X_i; \theta).$$

The maximum likelihood estimator is

$$\hat{\theta} = \arg \max_{\theta} L(\theta)$$

The **method of moments** estimates unknown parameters by equating sample moments with corresponding population moments.

For example, the first sample moment is

$$\frac{1}{n} \sum_{i=1}^n X_i,$$

which is equated with the first population moment

$$E[X].$$

Non-Parametric Model Estimation

A **non-parametric model** does not assume that the population distribution belongs to a specific probability distribution.

It makes fewer assumptions about the underlying population and estimates the distribution or its characteristics directly from the sample data.

The main idea is to allow the data to determine the shape of the underlying distribution.

- Empirical Distribution Function (EDF)
- Kernel Density Estimation (KDE)
- Rank-based estimation
- Median and quantile estimation

The **Empirical Distribution Function (EDF)** estimates the cumulative distribution function directly from the observed sample.

Non-Parametric Model Estimation

For a sample $X_1, X_2, \dots, X_n$, it is defined as

$$F_n(x) = \frac{1}{n} \sum_{i=1}^n I(X_i \leq x)$$

where $I(X_i \leq x)$ is the indicator function:

$$I(X_i \leq x) = \begin{cases} 1, & X_i \leq x \\ 0, & X_i > x \end{cases}$$

The EDF gives the proportion of observations that are less than or equal to x .
Consider the sample

2, 4, 5, 7, 9

For $x = 5$, three observations are less than or equal to 5.

Non-Parametric Model Estimation

Therefore,

$$F_n(5) = \frac{3}{5} = 0.6$$

Hence, approximately 60% of the observations are less than or equal to 5.

Kernel Density Estimation (KDE) is used to estimate the probability density function without assuming a particular distribution.

$$\hat{f}_h(x) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x - X_i}{h}\right)$$

where h is the bandwidth and K is the kernel function.

Non-parametric methods often use the **ranks** of observations instead of their actual numerical values.

Non-Parametric Model Estimation

For example, consider

25, 10, 40, 15

After arranging the observations:

10, 15, 25, 40

Their ranks are:

1, 2, 3, 4

Rank-based methods are useful when the underlying distribution is unknown or when the assumptions of parametric methods are not satisfied.

- Sign Test
- Wilcoxon Signed-Rank Test
- Mann–Whitney U Test
- Kruskal–Wallis Test
- Spearman's Rank Correlation

Parametric vs Non-Parametric Estimation

Parametric	Non-Parametric
Assumes a specific distribution	Makes fewer distributional assumptions
Uses a fixed number of parameters	More flexible estimation
Example: Normal distribution	Example: Empirical distribution
MLE and Method of Moments	KDE and rank-based methods
Less flexible	More flexible
Efficient when assumptions are correct	More robust to distributional assumptions

Conclusion

Descriptive statistics provides methods for organizing, summarizing, and presenting data, while frequentist statistics provides methods for making statistical inferences about populations using sample data.

The important descriptive statistical concepts include histograms, sample mean, sample variance, order statistics, covariance and covariance matrices.

Frequentist concepts include sampling, mean square error, consistency, confidence intervals, and parametric and non-parametric estimation.

Together, these concepts form an important foundation for statistical data analysis and statistical inference.

Thank You!